Great-Circle Distance on a Sphere

A short worked note

Ship of Tools — preview sample

Abstract

The shortest path between two points on a sphere follows a *great circle*. This note collects the two standard formulae — the spherical law of cosines and the numerically stable haversine — and works a small four-waypoint route as an example. It doubles as a preview fixture, exercising multi-page layout, display mathematics, a vector figure, and a table.

1 Setup

Let two points on a sphere of radius R have latitudes ϕ_1, ϕ_2 and longitudes λ_1, λ_2, and write the differences

$$\Delta\phi = \phi_2 - \phi_1, \qquad \Delta\lambda = \lambda_2 - \lambda_1.$$

The central angle θ subtended at the sphere's centre fixes the arc length through $d = R\,\theta$; everything below is a way to get θ from the coordinates.

2 The central angle

The spherical law of cosines gives it directly,

$$\cos\theta = \sin\phi_1 \sin\phi_2 + \cos\phi_1 \cos\phi_2 \cos\Delta\lambda,$$

but for nearly-coincident points this form loses precision to floating-point cancellation. The *haversine* identity avoids it:

$$\operatorname{hav}(\theta) = \operatorname{hav}(\Delta\phi) + \cos\phi_1 \cos\phi_2 \operatorname{hav}(\Delta\lambda), \qquad \operatorname{hav}(x) = \sin^2\tfrac{x}{2},$$

which rearranges to the working formula

$$\boxed{d = 2R \arcsin\sqrt{\sin^2\frac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2\frac{\Delta\lambda}{2}}}.$$

3 Geometry

Figure 1 fixes the picture: the great-circle arc (green) is the distance we want, $d = R\theta$; the straight chord (dashed) is the shorter Euclidean distance through the interior and is *not* what a traveller on the surface covers.

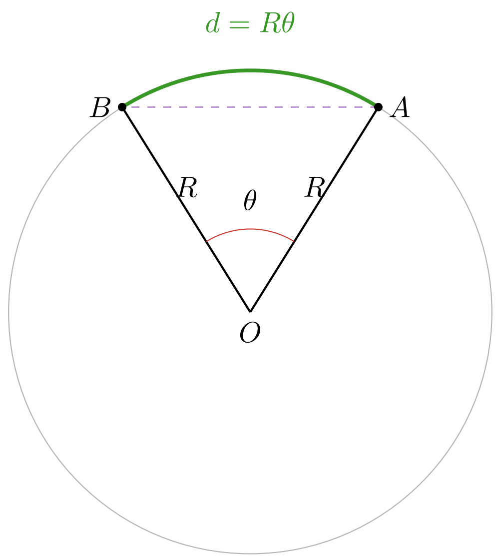

Figure 1: Two surface points A, B on a sphere of radius R. The great-circle distance is the arc $d = R\theta$, with θ the central angle at O; the dashed chord is the shorter straight-line distance.

Because $\theta \leq \pi$ for the minor arc, arcsin returns the correct branch and the haversine formula needs no quadrant bookkeeping — one reason it is the default in practice.

4 A worked route

Take a four-waypoint route across northern New Mexico and sum the legs on a spherical Earth, $R = 6371$ km. Table 1 lists each leg from the working formula above.

From	To	Leg (km)
Albuquerque	Santa Fe	93.5
Santa Fe	Black Mesa	23.7
Black Mesa	Taos	75.5
Total		**192.7**

Table 1: Per-leg great-circle distances for the demo route; the total, 192.7 km, is what `route_length` returns in the demo project.

The same computation lives in the demo project as `route_length`, dispatched over a vector of `Waypoints` — the figure it plots and this table are two views of one function.

References: Sinnott, R. W. (1984), *Virtues of the Haversine*, Sky & Telescope **68**(2), 159. — Vincenty, T. (1975), *Direct and inverse solutions of geodesics on the ellipsoid*.